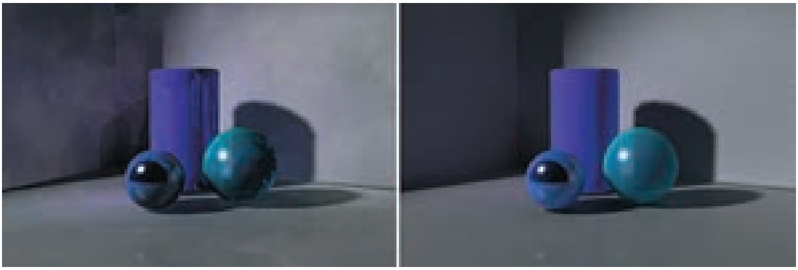
At low intensities, photons don't do much to provide additional illumination. Increasing the photon intensity adds diffusion to the scene. At very high intensities, the diffuse light from photons begin to overwhelm the direct light in the scene. With the direct light's intensity turned down to 0, indirect light created strictly by the scene's photons can be seen.

5.6.4 Final Gathering

Final Gathering is used as a finishing tool for Global illumination and can help reduce the need for high numbers of photons. In its simplest sense, it can be used as a blending algorithm, and it smoothens and interpolates the light created by the photons to simulate a radiosity solution.

Final Gathering is similar to raytracing. While raytracing traces light rays from the camera's point of view, Global Illumination starts at the light source and traces from there. Final Gathering rays are emitted from a light; when they hit a surface, mental ray calculates the way the rays are scattered, along



Global Illumination photons are plainly visible in this scene (left); many more photons could be added, but at the cost of rendering time. Final Gathering smoothens out the graininess in this scene (right), creating a good final result.

with their new energies. This process is continued, and the secondary rays of light continue to bounce off other surfaces and so on, creating a soft and realistic lighting of the scene.

5.6.5 Caustics

Caustics simulate the way light refracts through a complex surface or reflects off it. Caustics are useful in simulating such surfaces as water, glass, and metals. Working with caustics can be slightly tricky because several variables affect the way the final render looks. The photon intensity, the exponent of the falloff, distance of light source, number of photons, and the materials of the objects all affect the outcome.